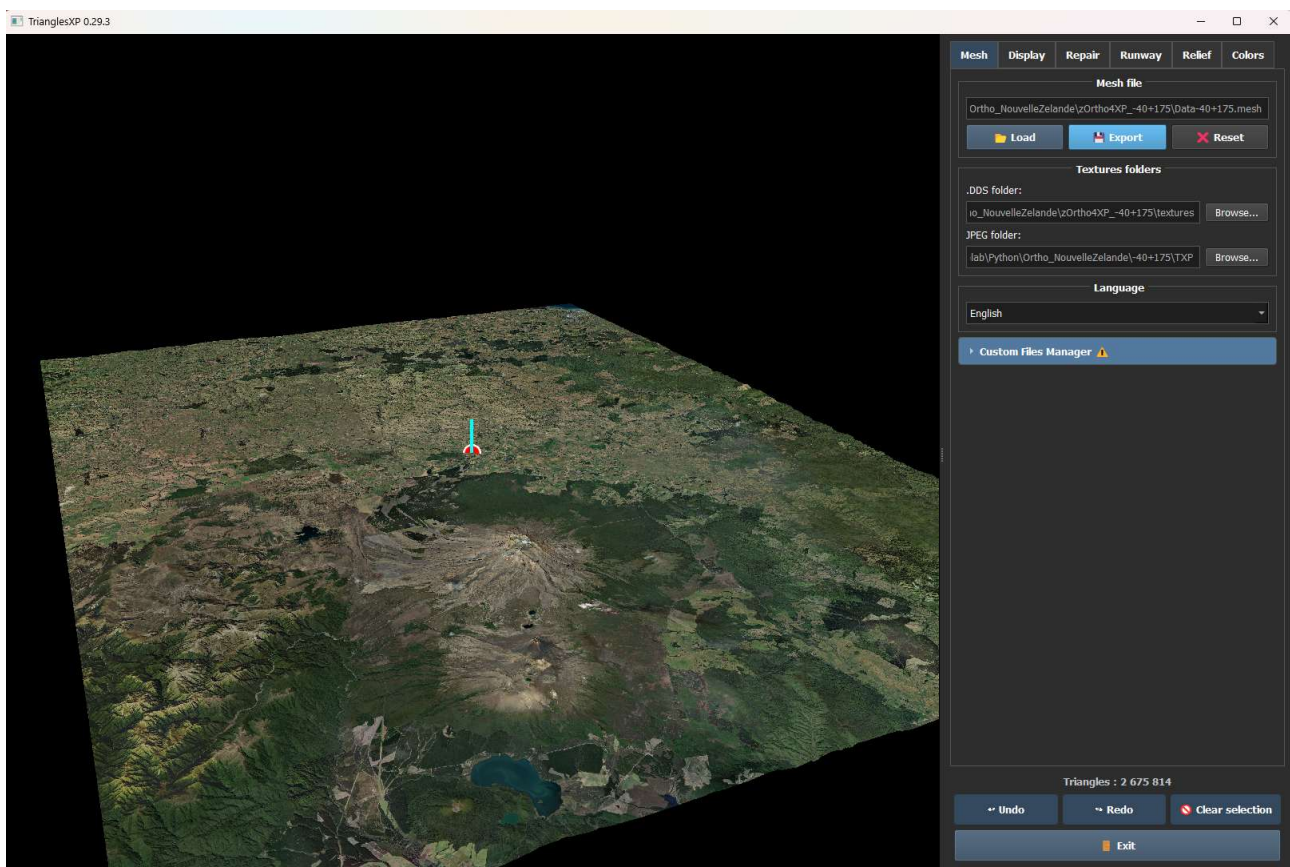


TrianglesXP

User Manual



Author : stef2us
Version 0.29.3
2026-07-23

Table of contents

1. Introduction.....	3
2. Mesh tab.....	4
3. Display tab.....	5
4. Basic moves and actions.....	6
5. Repair tab.....	8
6. Runway tab.....	13
7. Relief tab.....	15
8. Colors tab.....	19
8.1 Selection.....	21
8.2 Color adjustment.....	22
8.3 Batch mode.....	23
8.4 Color Layer.....	24
8.5 Export Orthophotos.....	25
9. Post-actions in Ortho4XP.....	27
9.1 Only mesh was altered.....	27
9.2 Only orthophotos were altered.....	28
9.3 Mesh and orthophotos were altered.....	30
10. File management.....	31
10.1 Custom Json files.....	31
10.2 Settings.ini.....	32

1. Introduction

TrianglesXP is a tool written in Python.

It was created for modifying ortho tiles made with Ortho4XP.

2 types of modifications can be done :

#1 : alter the mesh

and/or

#2 : alter the satellite pictures colors.

The workflow is the following :

1. create a tile with Ortho4XP
2. run TrianglesXP, open the .mesh file
3. (#1) make changes to the mesh
4. (#1) export the mesh to a new .mesh file
5. (#2) make color changes to user defined selections
6. (#2) save the changes to a new set of .JPG files
7. rerun required steps in ortho4xp to apply the changes on the tile

Note : I used shred86's fork of Ortho4XP, versions 1.40.11 to 1.40.13, to develop and test TrianglesXP.

2. Mesh tab

Mesh | Display | Repair | Runway | Relief | Colors

Mesh file

Ortho_NouvelleZelande\zOrtho4XP_-40+175\Data-40+175.mesh

Load Export Reset

Textures folders

.DDS folder:
io_NouvelleZelande\zOrtho4XP_-40+175\textures Browse...

JPEG folder:
lab\Python\Ortho_NouvelleZelande_-40+175\TXP Browse...

Language

English

Custom Files Manager ⚠

File :
custom_flat.json

Entry :
Wuhapehu_lake_01

Refresh Rename Delete

Load : open a mesh file.

Export : save to a new mesh file after you made changes

Reset : cancel all changes made to the mesh

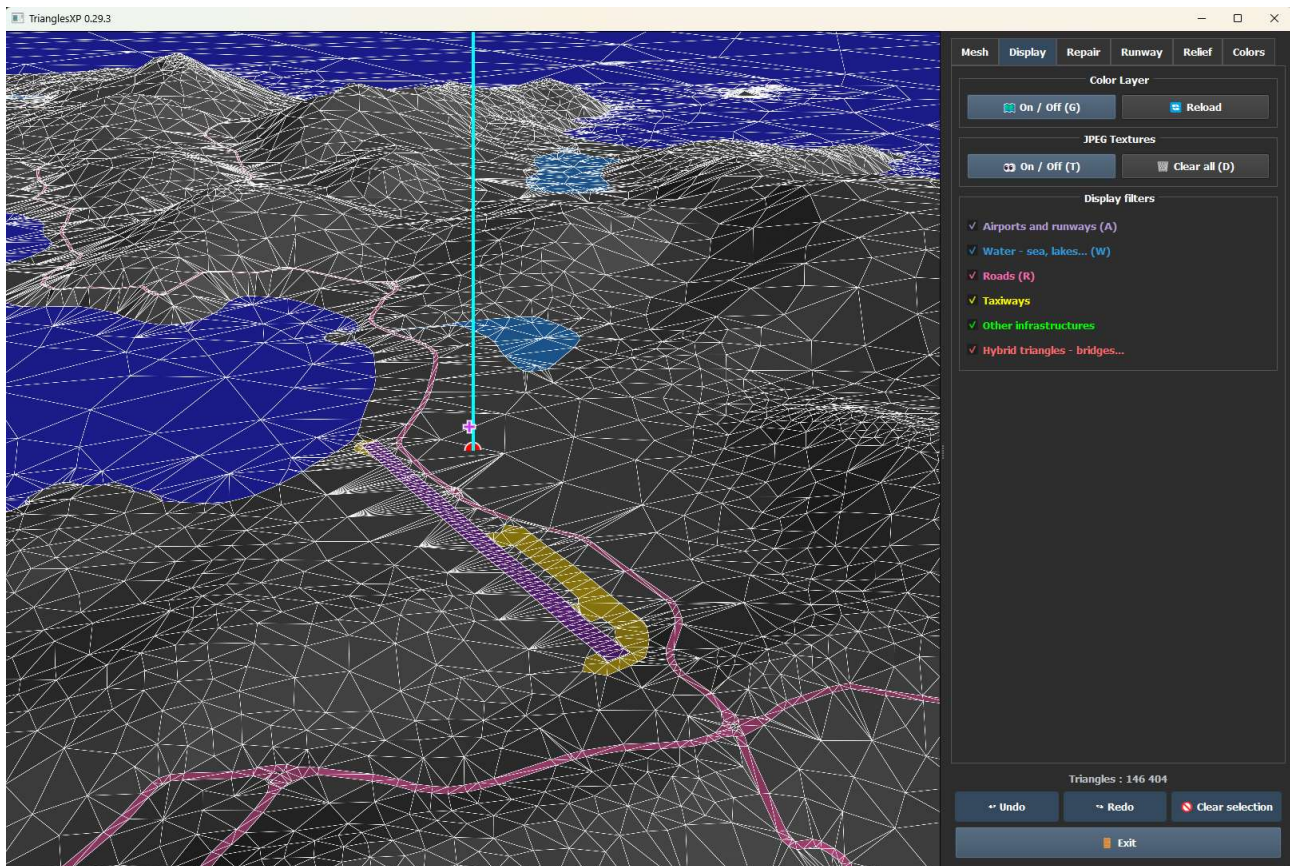
.DDS folder : filled automatically after mesh load

JPEG folder : not mandatory unless you want to display the satellite texture, or you want to modify the colors. Look in internal / ortho4xpdata / orthophotos / .

Language : restart TrianglesXP to take account of change.

Custom files manager : allows to delete or rename items created in TrianglesXP more safely than modifying the .JSON files yourself.

3. Display tab



Color layer :

On / off : (shortcut G) displays/hide downgraded satellite picture over the whole tile

Reload : to be used after you changed « JPEG folder » in the mesh tab

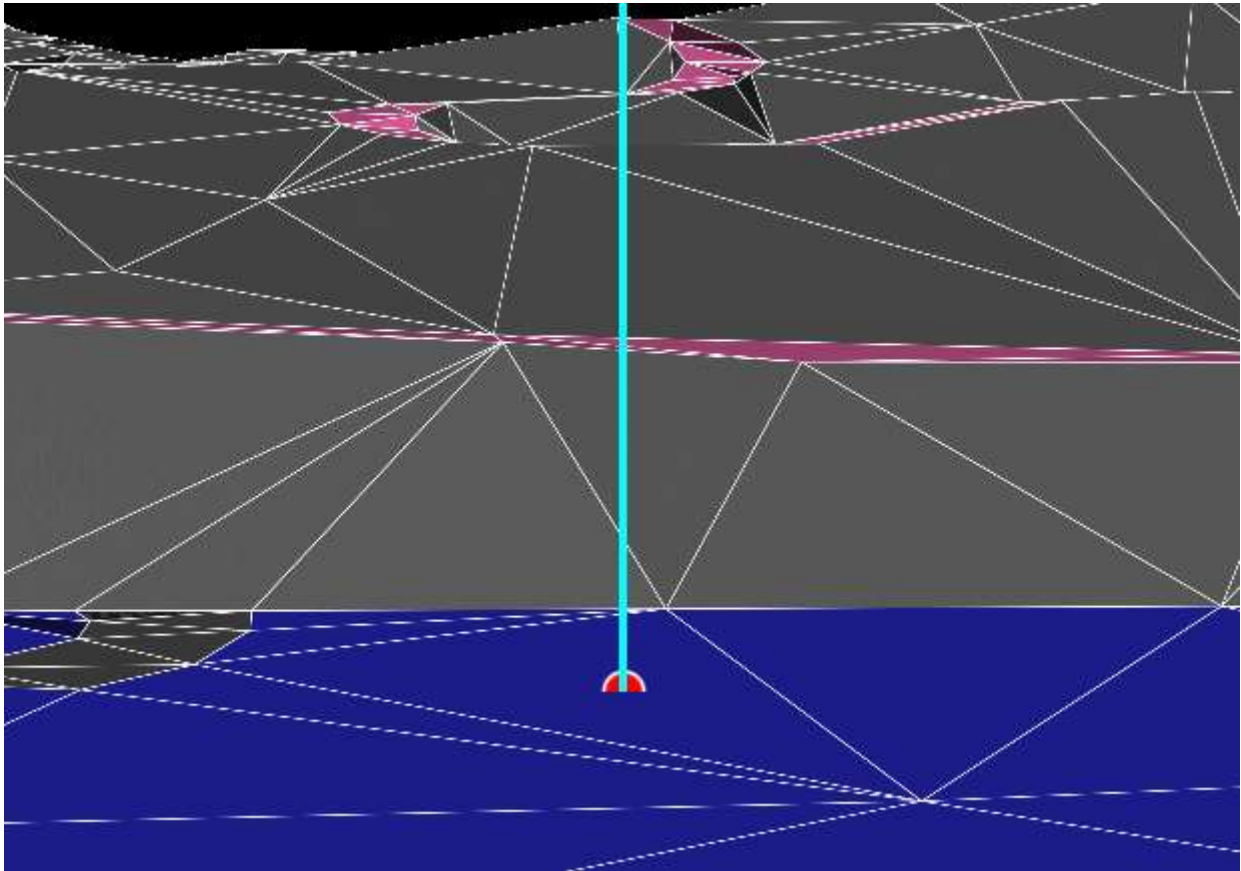
JPEG textures :

On / off : (shortcut T) displays/hide original satellite picture in the area around the camera pivot (red ball with cyan vertical line). Up to 4 jpegs can be displayed simultaneously.

Clear all : (shortcut D) erases all the HD textures currently displayed

Display filters : make special triangle types appear differently on the mesh.

4. Basic moves and actions



Moves :

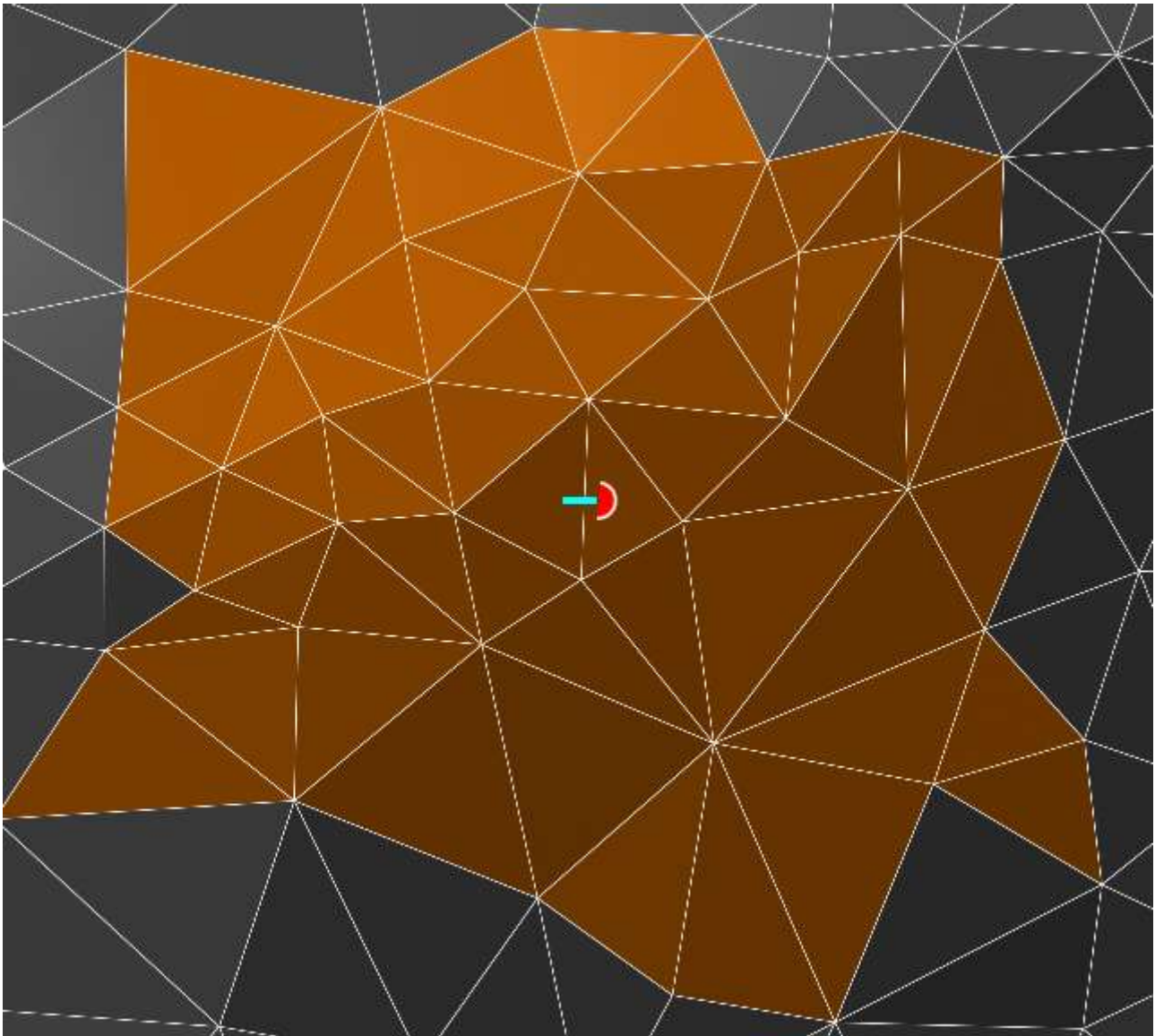
Camera pivot : the red ball and the cyan vertical line will always remain at the center of the view.

Navigate on the mesh either with the keyboard arrow keys, or with SHIFT + drag.

Change view axis : mouse drag.

Zoom in / out : mouse wheel.

Change perspective (avoid) : SHIT + right drag.



Actions : to be done with vertical point of view.

Create a selection : CTRL + click then drag to draw a rectangle. All the triangles crossed by the rectangle will be included in the selection.

Delete a selection : type ESC or click on « Clear selection » button (bottom right).

Add a triangle to a selection : CTRL+click in the triangle.

Remove a triangle from a selection : CTRL + right click in the triangle.

Add multiple triangles to a selection : draw another selection. Will be added to the previous one.

Remove multiple triangles from a selection : draw a selection with CTRL + right click + drag.

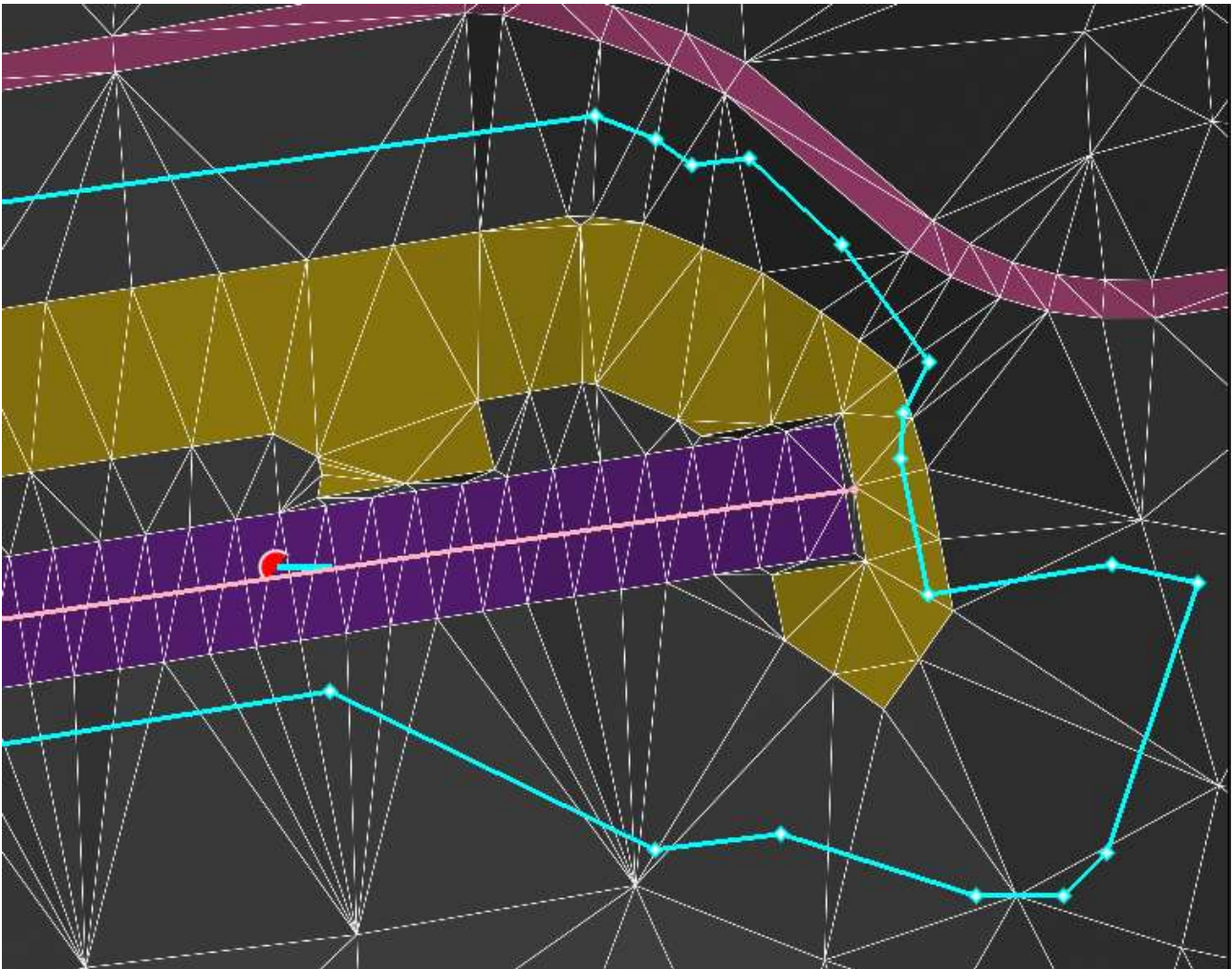
5. Repair tab

The image shows a software interface with a top navigation bar containing tabs: Mesh, Display, Repair, Runway, Relief, and Colors. The 'Repair' tab is selected and active.

Below the tabs, the interface is organized into several sections:

- Flatten 1**:
 - A text field for 'Current name:'.
 - A green button labeled 'Enable Area Trace'.
 - Two buttons: 'Load' (with a folder icon) and 'Save' (with a save icon).
 - Surface parameters**:
 - A checked checkbox for 'Convert to road type'.
 - A text field for 'Side width (m):' with the value '100'.
 - A 'Precision' slider set to '1'.
 - Polygon post-edition**: Three buttons labeled 'Edit', 'Insert', and 'Delete'.
 - Tilted surface**: A toggle switch is turned off, and a 'Disabled' button is visible.
 - Four sliders for 'Altitude point A', 'Rotation A (°)', 'Altitude point B', and 'Rotation B (°)', each with a corresponding numerical value field (all showing '0,00 m' or '0,00').
 - Two buttons: 'Validate polygon' (with a green checkmark) and 'Clear polygon' (with a red X).
- Flatten 2**:
 - A 'Target Z:' slider set to '0,0 m'.
 - Two buttons: 'Apply' (with a green checkmark) and 'Cancel' (with a red X).
- Two expandable sections at the bottom: 'Vertices' and 'Triangle Attributes', each with a right-pointing arrow.

This feature allows to draw a polygon on the mesh, and turn it into a flattened surface that will be horizontal or tilted.



Enable area trace : turns on polygon draw mode. Draw the polygon point by point with CTRL+click. Close the polygon with CTRL + right click. You can undo / redo when drawing with either the buttons at the bottom, or CTRL+Z / CTRL+Y.

Load : load a polygon that was previously saved. You can create and save as many polygons as you want without being obliged to save a mesh file for each attempt.

Save : save the definition of a polygon into a .JSON file.

Surface parameters :

Convert to road type : this is what is done for flattened surfaces in OSM patches. Not mandatory.

Side width : width in meters of the embankments around the surface. These are made to have smooth terrain transitions.

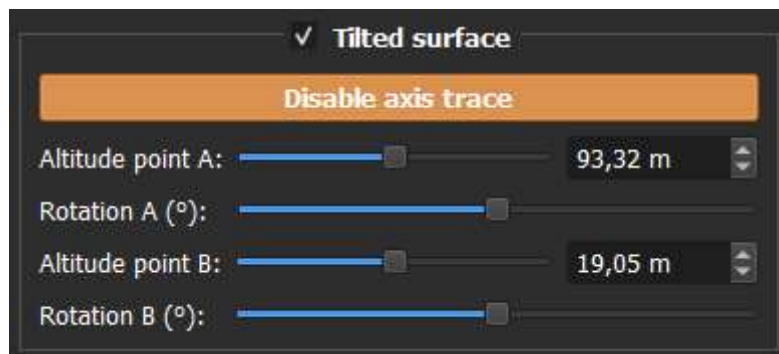
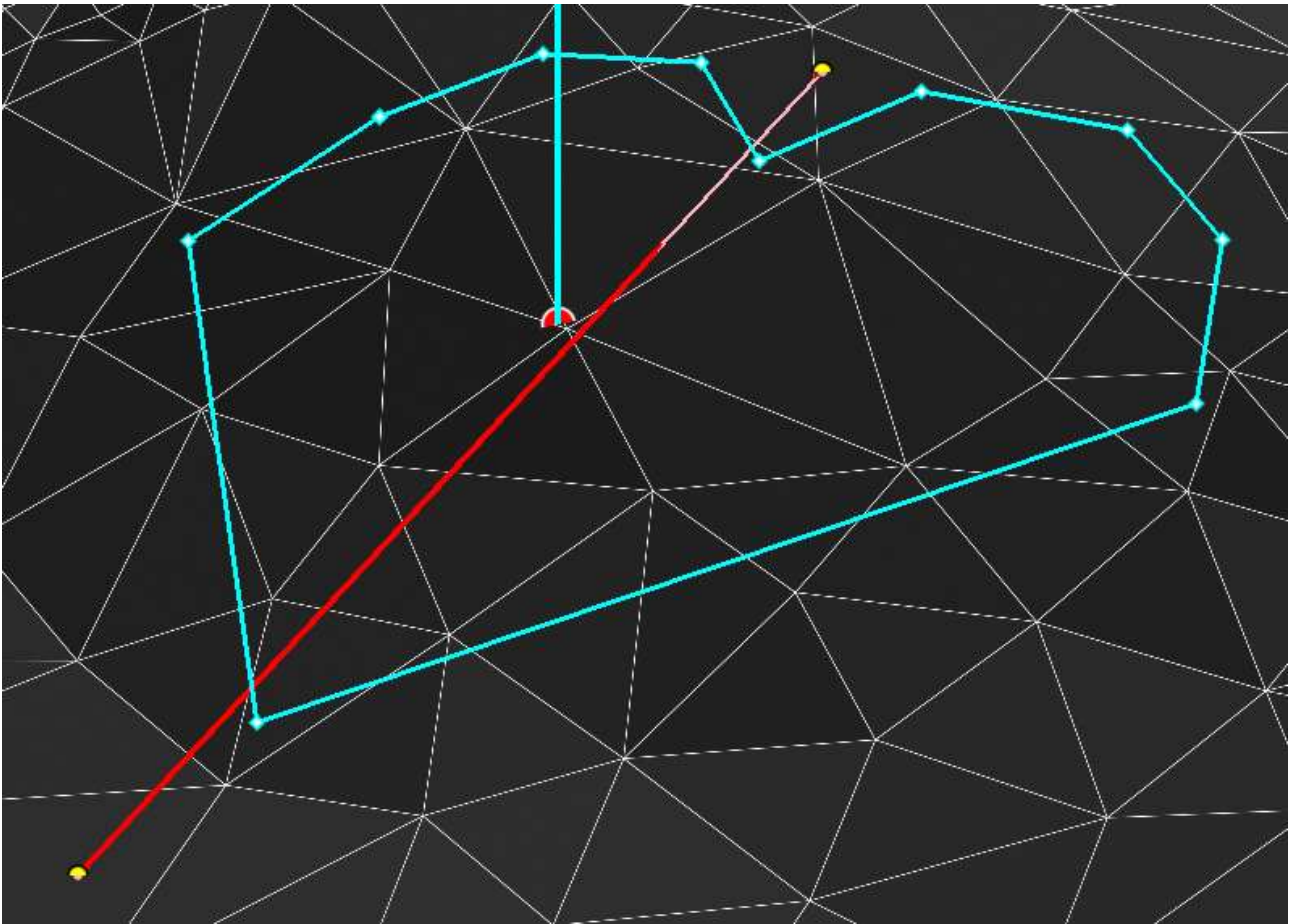
Precision : number of triangles subdivisions to achieve the process. The precision can increase automatically if the configured number is too low to achieve the surface creation.

Polygon post-edition :

Edit : click button. CTRL+click close to an existing point. Now CTRL+click on the new location for this point. The polygon will have its border updated.

Insert : click button. CTRL+click close to an edge of the polygon. A new point is created on it. Now CTRL+click on the new location for this point. The polygon will have its border updated.

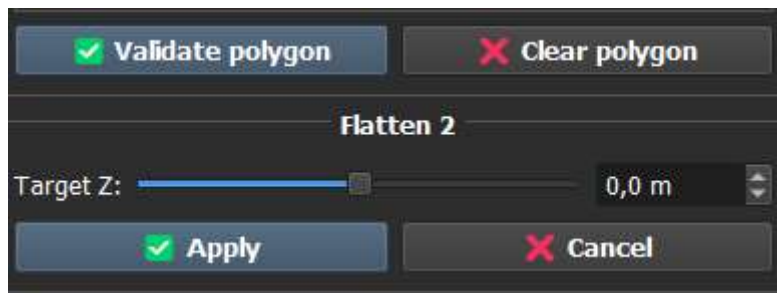
Delete : click button. CTRL+click close to an existing point. Confirm deletion. The polygon will have its border updated.



Tilted surface (optional) : enable it if you don't want the surface to be horizontal.

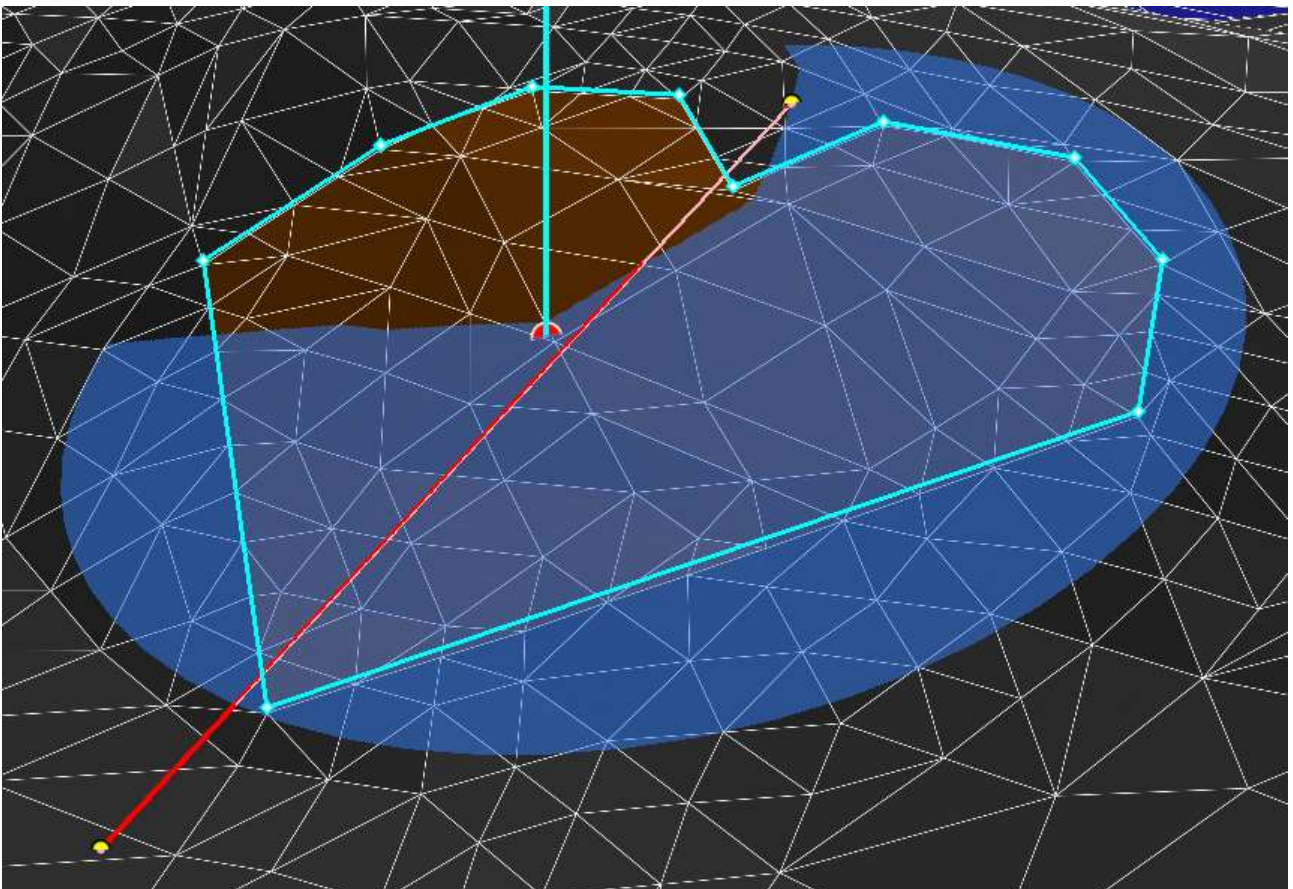
Enable axis trace (green => orange) : enables axis draw mode. Create 2 points with CTRL+click.

The axis is materialized with a red line between 2 yellow balls. You can use the sliders or the spin boxes to change their altitudes or make them rotate. Click on Disable axis trace or uncheck Tilted surface to delete the axis.



Validate polygon : attempts to build the polygon in the mesh. May fail, especially if the polygon is too small compared to surrounding triangles, or if there are water triangles close to the polygon. In that case, edit the polygon or draw a new one. You **MUST** save the polygon **BEFORE** you validate it.

Clear polygon : deletes current work.

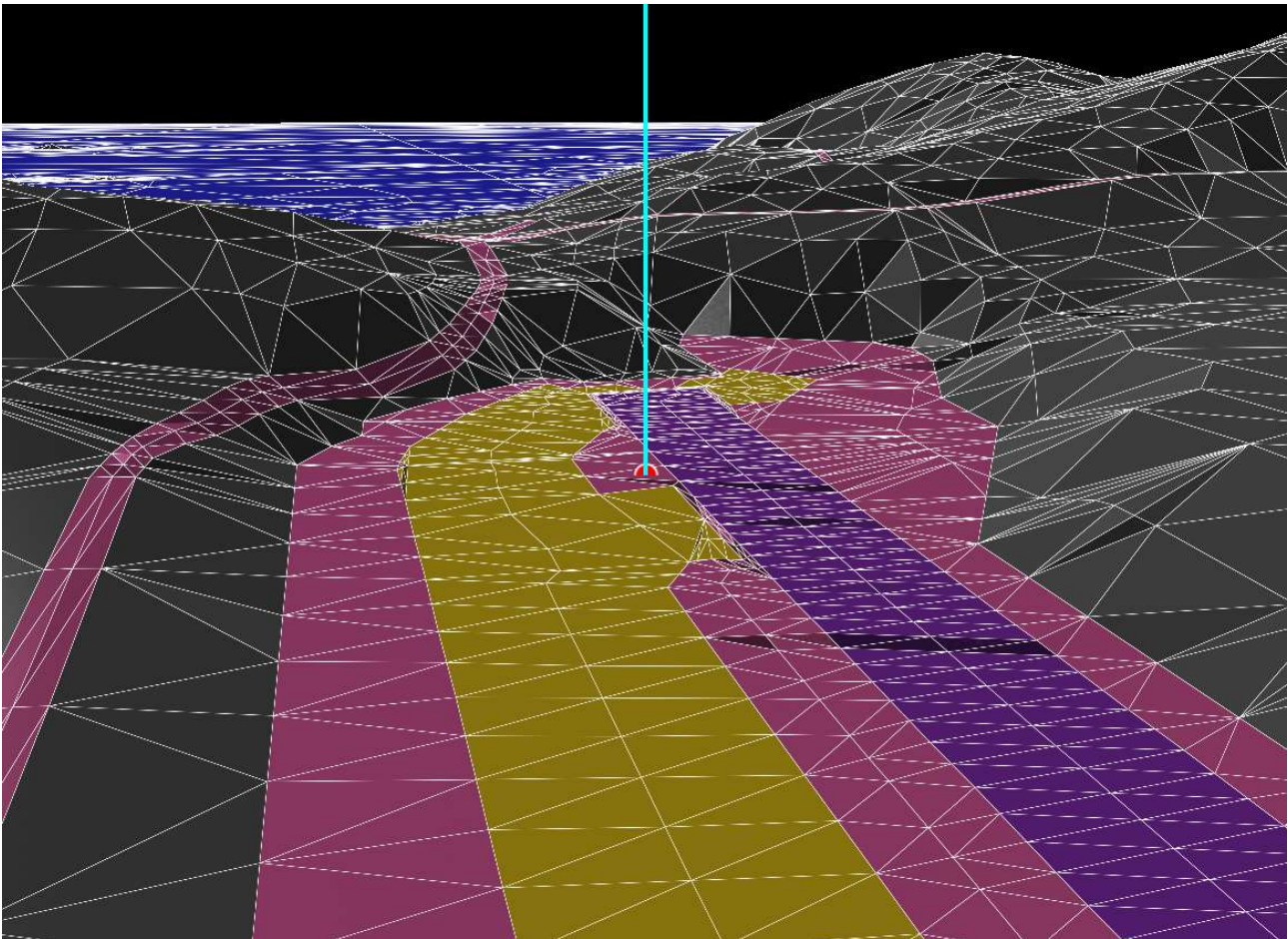


Flatten 2 : activates when the polygon is validated.

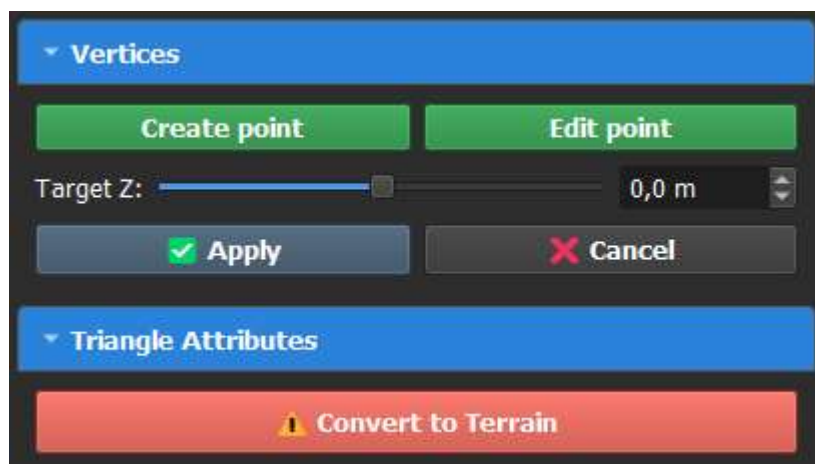
Target Z : enabled if you did not go for a tilted surface. In that case, you can define the final altitude of the flattened surface. Tip : always better to create a tilted surface with horizontal axis, so that the altitude is saved to the .JSON file.

Apply : creates the flattened surface.

Cancel : deletes current work.



Vertices : alter mesh directly.



Create point : creates a new point at the middle of the mesh triangle edge that is the closest to the camera pivot and turn it into Z-edit mode.

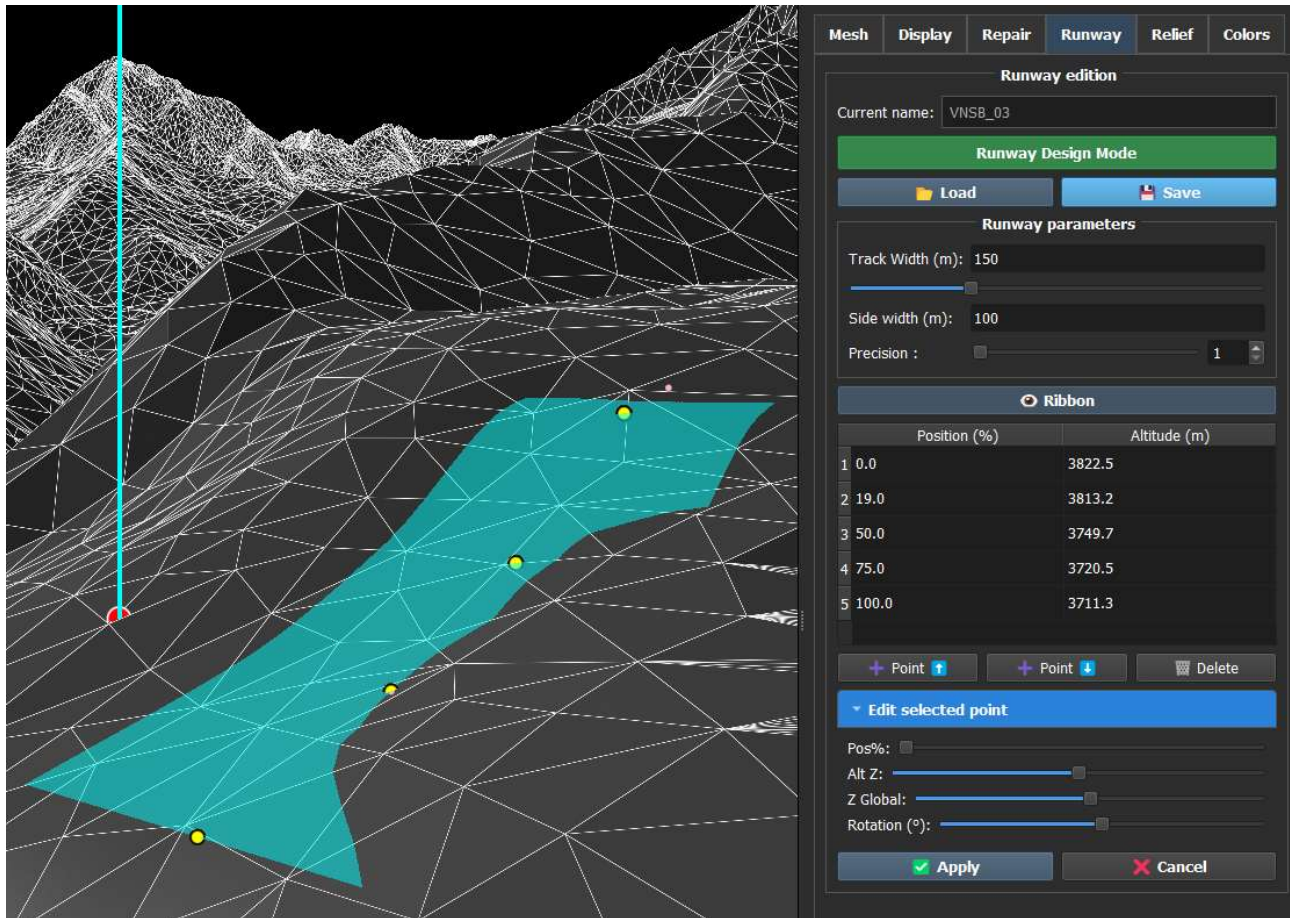
Edit point : select the closest point to the camera pivot and turn it into Z-edit mode.

Target Z : define new altitude for the point being created/edited/

Apply / **cancel** buttons to validate or not. Undo / redo are handled here.

Triangle attributes (should not be used) : converts triangles in selection to basic terrain type.

6. Runway tab



Set runway boundaries / Runway design mode : enable / disable runway design mode. Start with defining 2 points with CTRL + click. These points will be the boundaries of your runway. Use T shortcut to adjust their locations.

Load : load a runway that was saved into a .JSON file.

Save : create a backup of your work into a .JSON file.

Runway parameters :

Track width : width of the track. Think of including runway sides that must have same altitude profile as the runway itself.

Side width : width in meters of the embankments around the track. These are made to have smooth terrain transitions.

Precision : number of triangles subdivisions to achieve the process.

Line / Ribbon button : switch view between red line or transparent cyan ribbon.

Runway points table (up to 7 points) :

+ Point Up : create a new point above the selected one, unless it's the first in the list.

+ Point Down : create a new point under the selected one, unless it's the last in the list.

Delete : drop selected point.

Edit selected point : to make selected point move as desired along the runway axis.

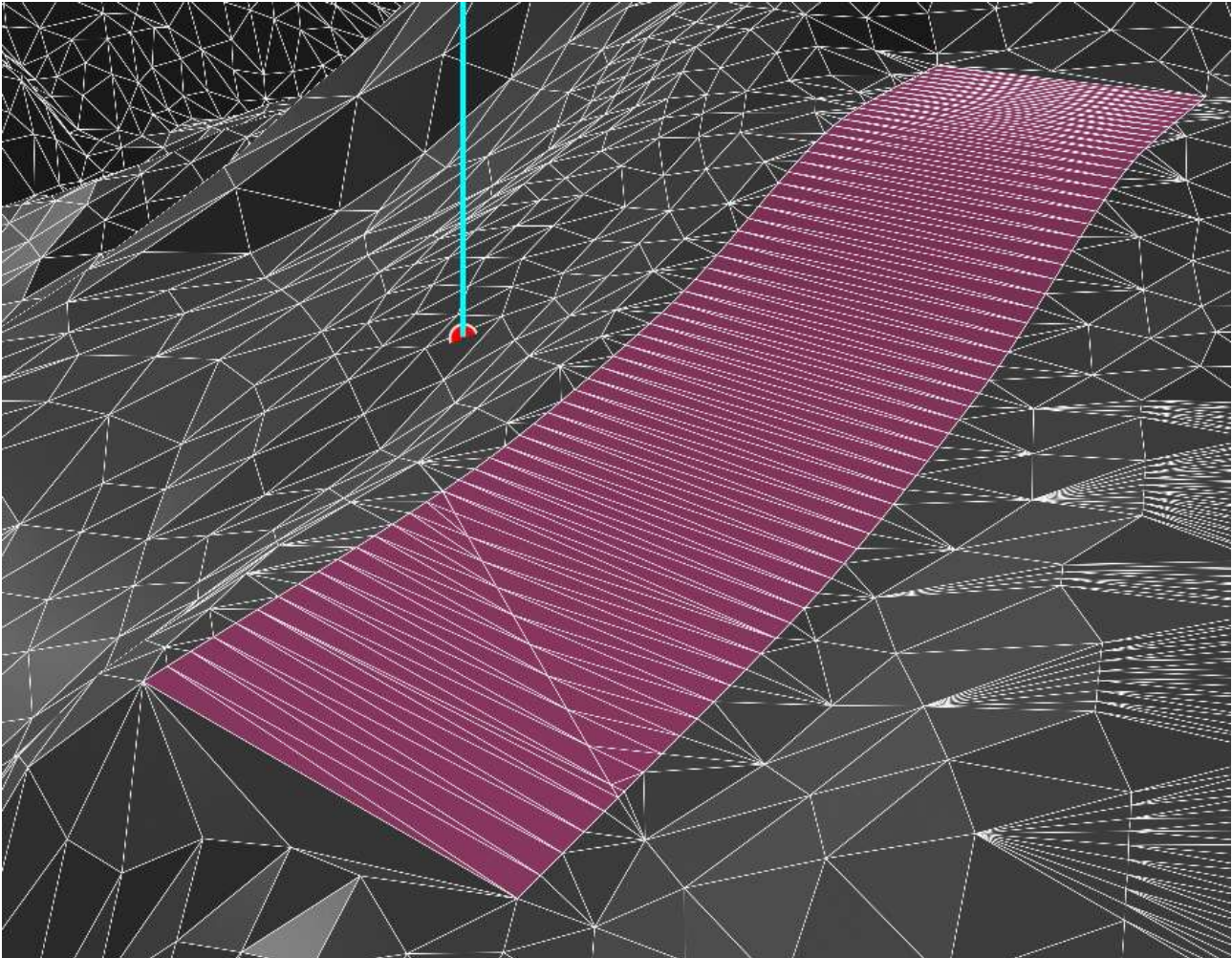
The axis shape will change if **Pos%** or **Alt Z** is changed.

Z global makes all the points move together.

Rotation applies when a runway boundary is selected in the table (first or last point).

Apply : create the runway in the mesh.

Cancel : remove your work.



Runways are created with road type triangles.

The layout with thin triangles is made on purpose to guarantee airplane stability at take off and landing.

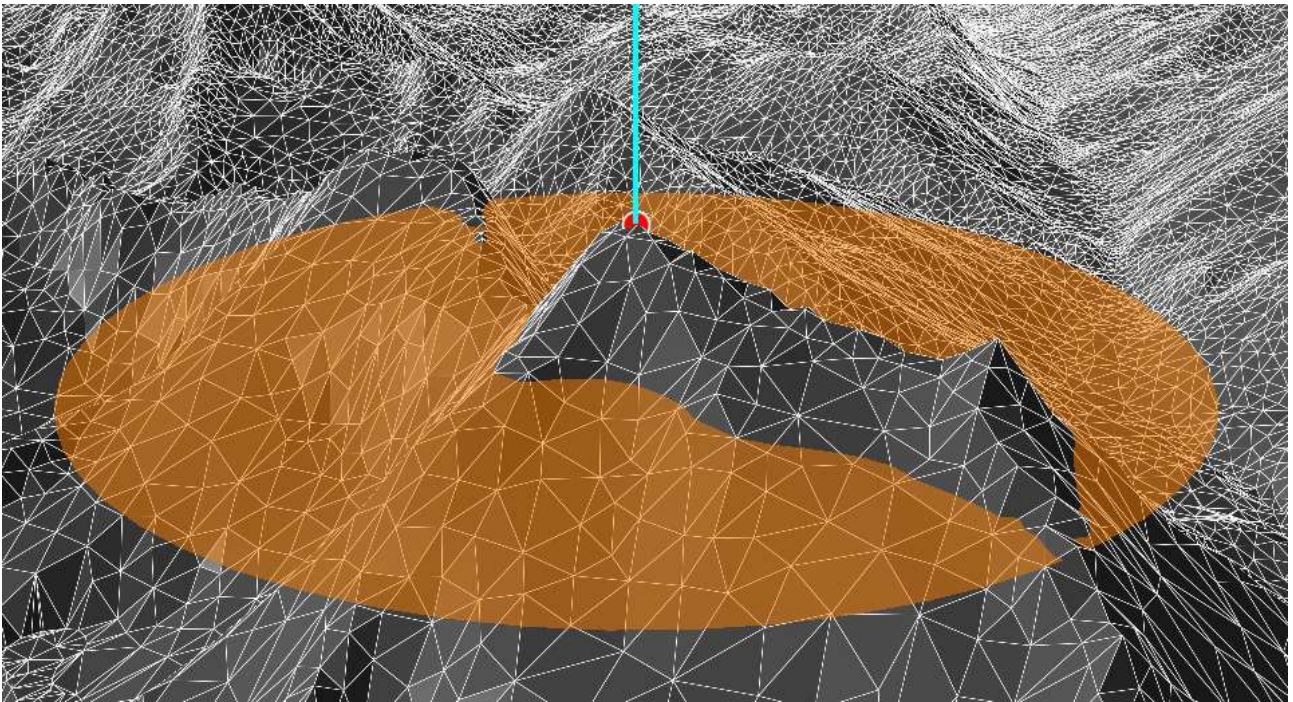
A runway with only 2 points will have a straight profile (most common case).

Increasing the precision and/or the side width will make the runway sides smoother.

7. Relief tab

The image shows a software interface with a top navigation bar containing tabs: Mesh, Display, Repair, Runway, Relief (selected), and Colors. Below the tabs, there is a section titled "Summit selection" with a green "Enable Selection" button. Under this button, there are input fields for "Alt Pivot (m): 3716.3", "Radius (m): 1000" (with a slider), and "Alt Base (m): 3516.3" (with a slider). At the bottom of this section are "Apply" and "Cancel" buttons. Below this is a section titled "Smoothing - Relief". It has a "Preset:" dropdown menu set to "Field". Below that is a "Smooth iterations" slider set to 1. A checkbox labeled "Chaos parameters" is checked. Under this checkbox are four parameters: "Precision:" (slider set to 0), "Octaves:" (slider set to 4), "Amplitude:" (slider set to 25), and "Scale:" (slider set to 150). At the bottom of the "Smoothing - Relief" section are three buttons: "Save", "Apply", and "Compare".

This part is mostly dedicated for smoothing terrain, especially at mountain tops. The « chaos » optional section allows to generate random bumps.



Summit selection : use this as an alternative to making a selection with CTRL + click + drag. Typically, move the camera pivot to the top of a mountain, then click **Enable selection**.

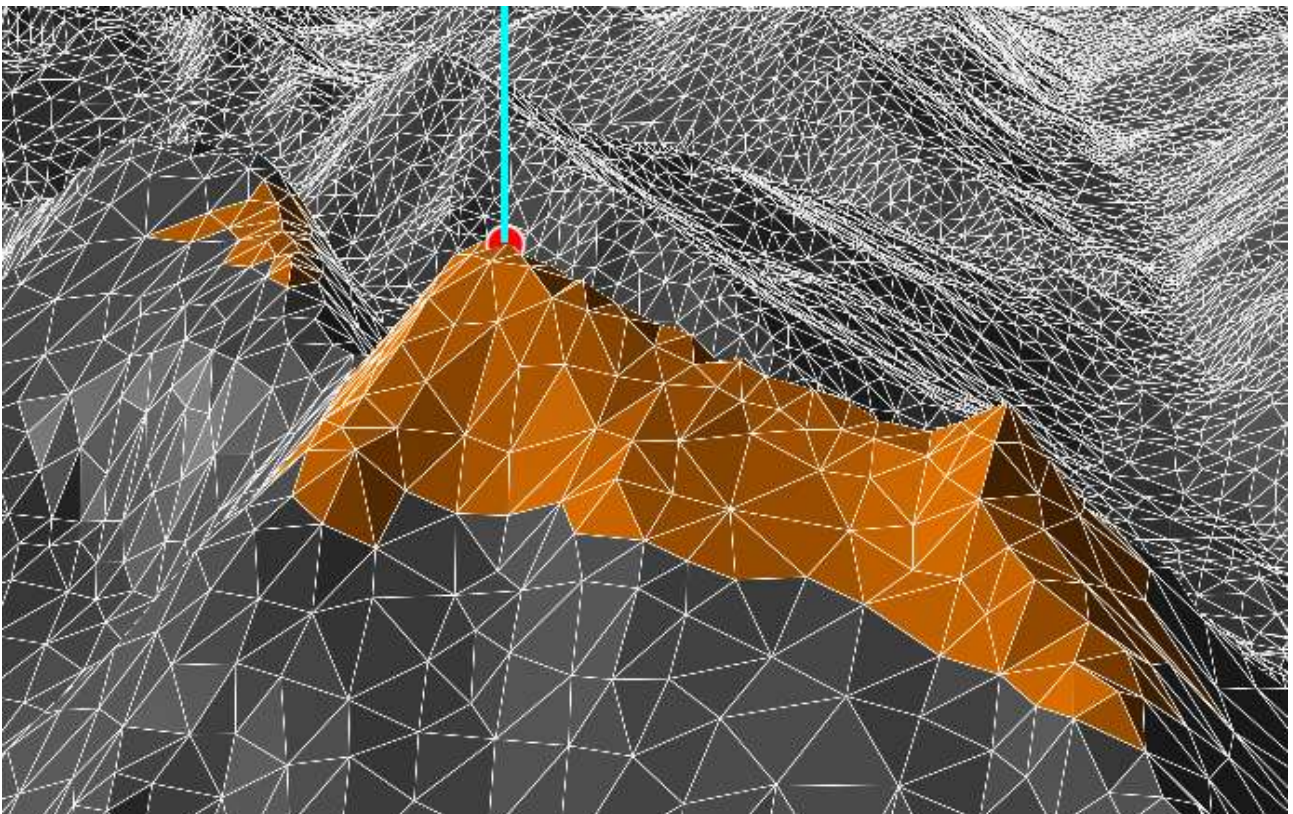
Alt pivot shows the current altitude of the red ball.

Radius is the radius of the orange translucent disk.

Alt base is the disk altitude offset under the pivot altitude.

Change **radius** and **Alt base** so that all the triangles above the disk will get selected.

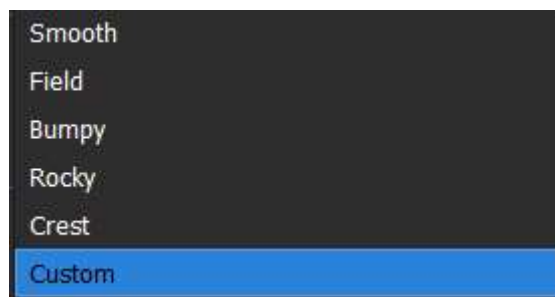
Apply / cancel : create the selection or not.



Smoothing – Relief section :



Presets :



Smooth : smooth terrain with no random chaos.

Field, Bumpy, Rocky, Crest : predefined random chaos presets.

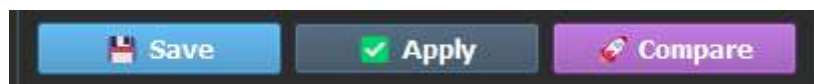
Custom : lets you choose chaos parameters.

Smooth iterations : number of subdivisions + smoothing loops.

Chaos parameters : check to enable (smoothing is mandatory before to apply chaos).

Precision : additional precision after smoothing step.

Octaves, Amplitude, Scale : chaos style and size adjustments.



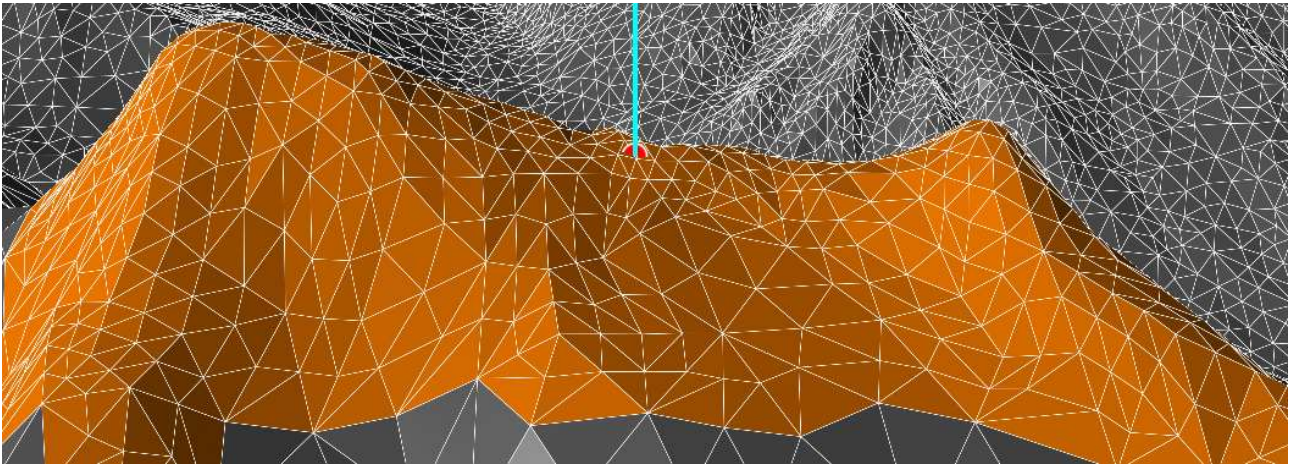
Save : if you played with parameters, the preset name has turned into « Custom » and you can save it with an alternate name.

Apply : apply the current smooth/chaos configuration to the mesh.

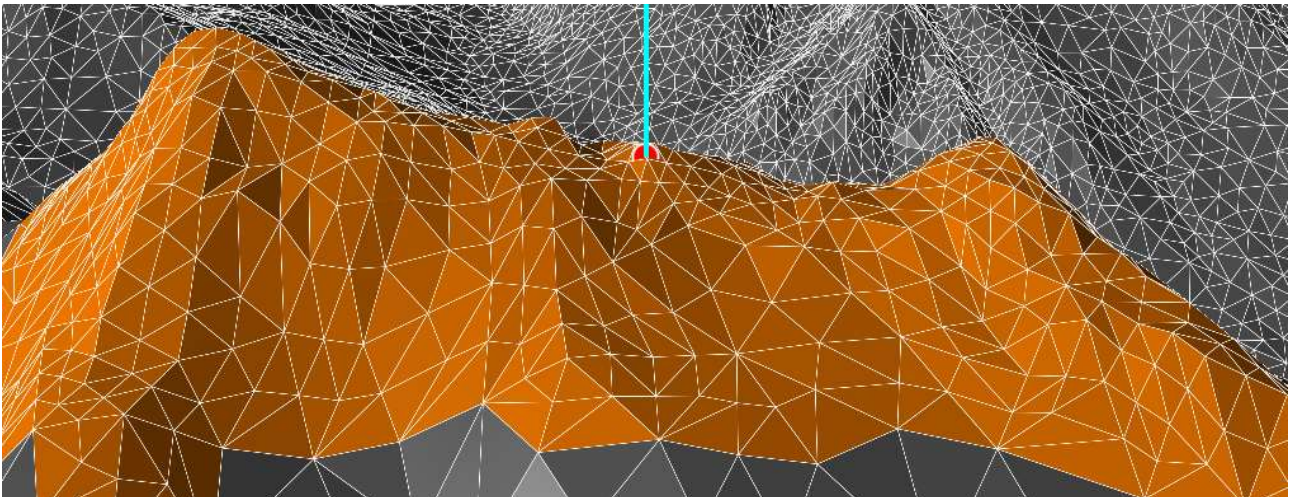
Compare : undo of the previously applied configuration and replace with the current one. Useful for accurate parameters adjustments.

Chaos presets examples :

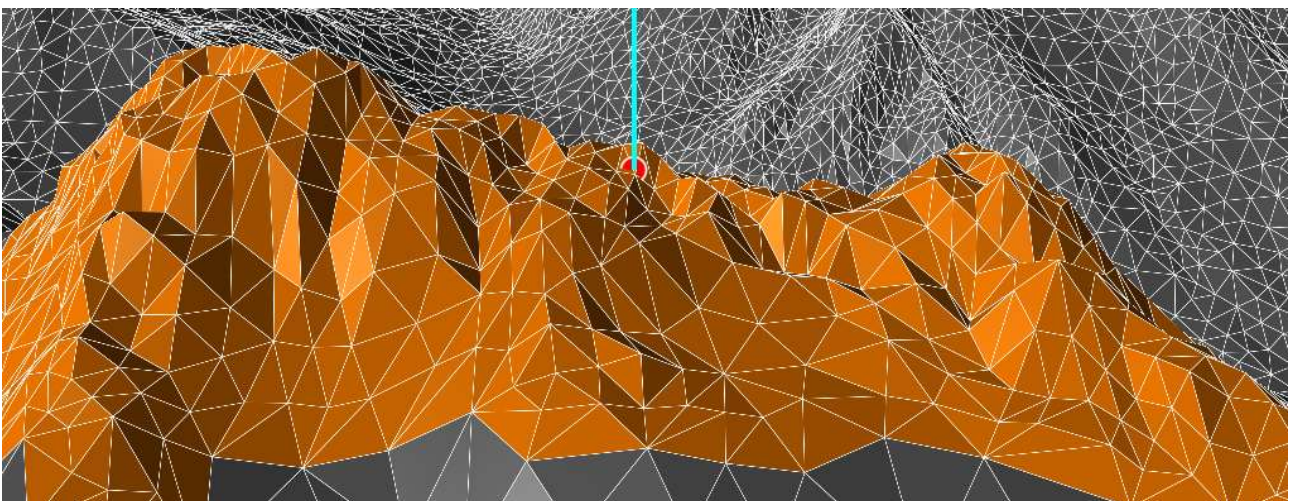
Field :



Bumpy :

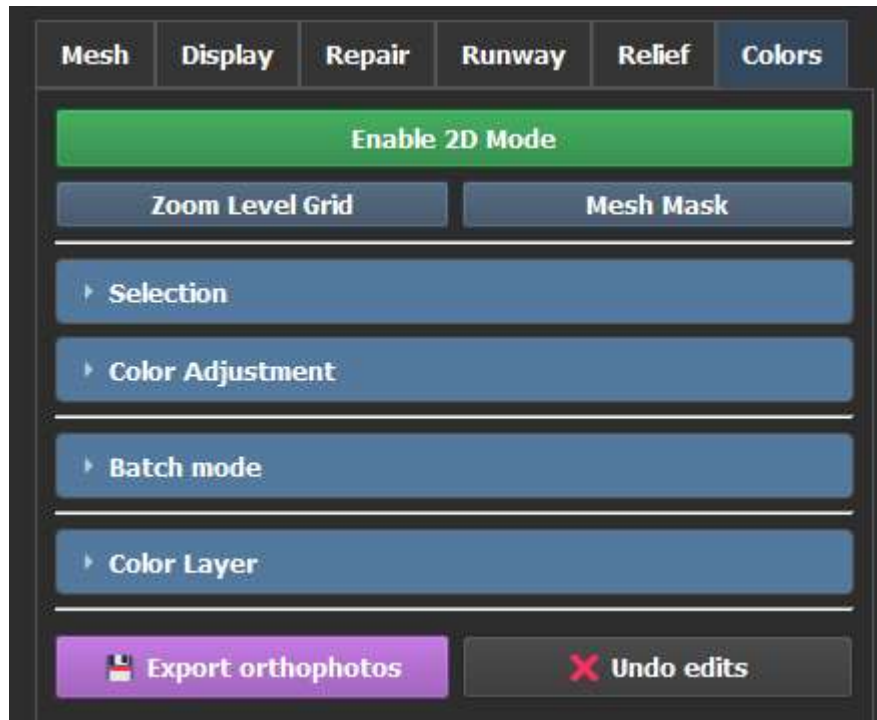


Crest :



8. Colors tab

This is the place where you can quite easily modify the satellite photos colors, either by yourself, or by applying selections & color presets made and shared by someone else.



Enable 2D mode : change view to 2D => **Disable 2D mode** : to get back to 3D (any work in progress is kept alive while switching mode).

To switch to 2D, the color layer must have been loaded before while in 3D mode (G shortcut). It means that the ortho tile must have been completed until the end in Ortho4XP, including .DDS and .JPG files in the « orthophotos » directory tree.

Navigate in the map with mouse drag.

Zoom in / out with mouse wheel.

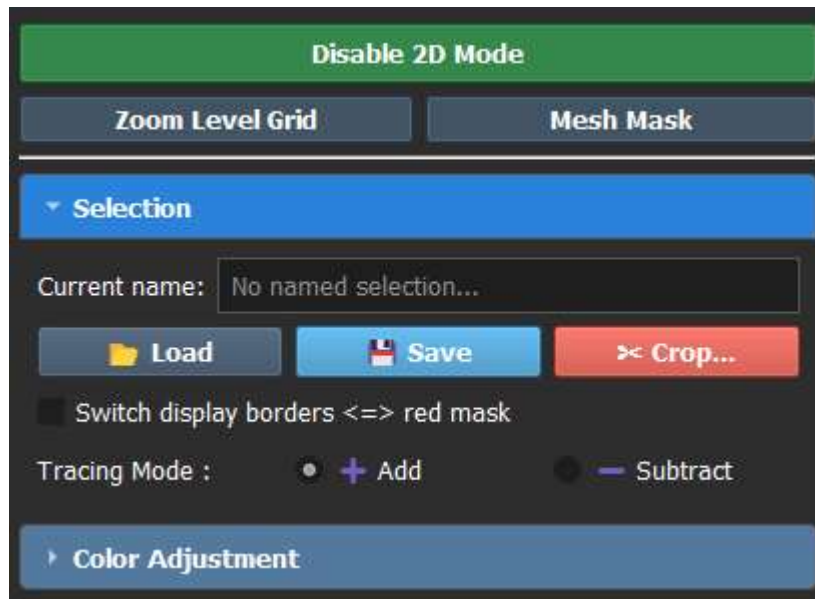
Zoom level grid : displays layout of the different ZL tiles created by Ortho4XP.

Mesh mask : hide the parts of the satellite pictures that are outside mesh limits.



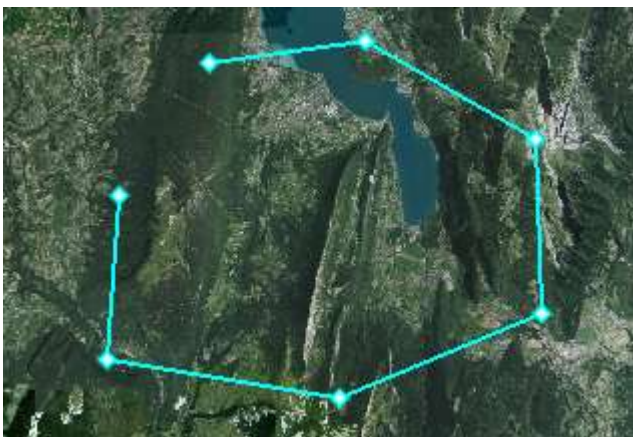
Example : -40+175 with ZL16, 17 and 18 jpeg pictures in the tile. Mesh mask on the sides.

8.1 Selection



Actions :

It is possible to draw a selection with SHIFT + click and close it with SHIFT + right click.



Otherwise, **load** a single selection. **Save** to backup the current selection to a .JSON file.

Crop... : crops current selection with another selection that was saved to disk.

Switch display borders <=> red mask : display mode of current selection.

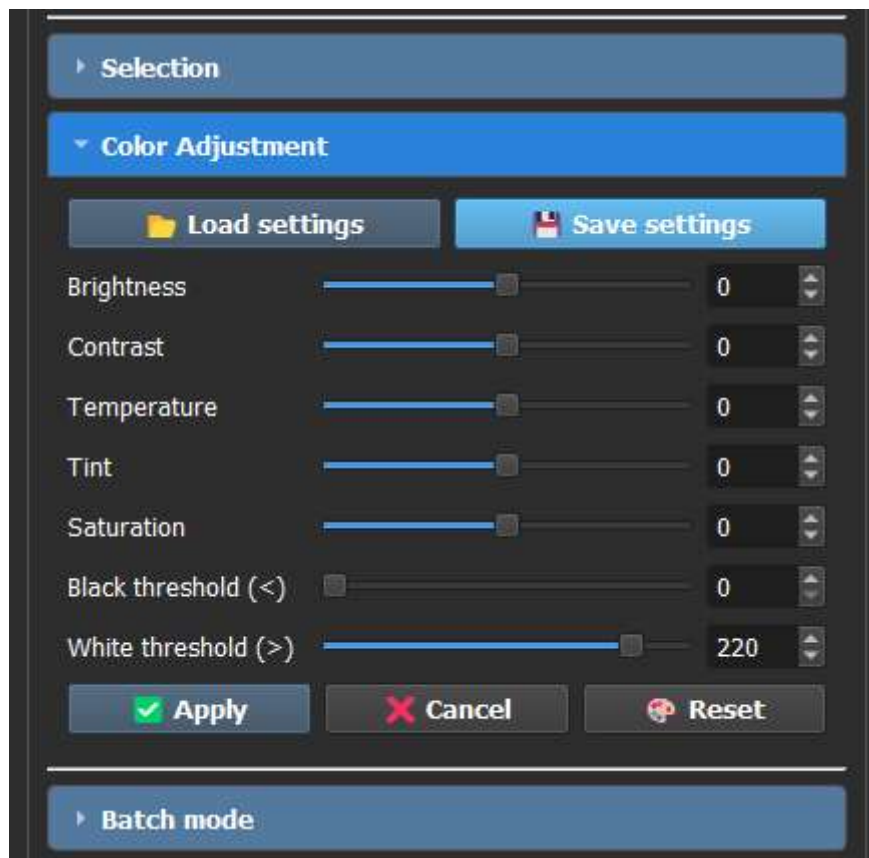
Tracing mode :

« **Add** » : multiple polygons can be drawn and finally be considered as 1 selection. Also possible to enlarge an existing polygon this way.

« **Subtract** » : crop a polygon with a new one that you draw instead of using a saved selection by using the crop button.

8.2 Color adjustment

Applies to the current selection only. To apply to whole tile, draw a rectangle around the map.



Load settings : load parameters if previously saved for current selection.

Save settings : backup your work to a .JSON file.

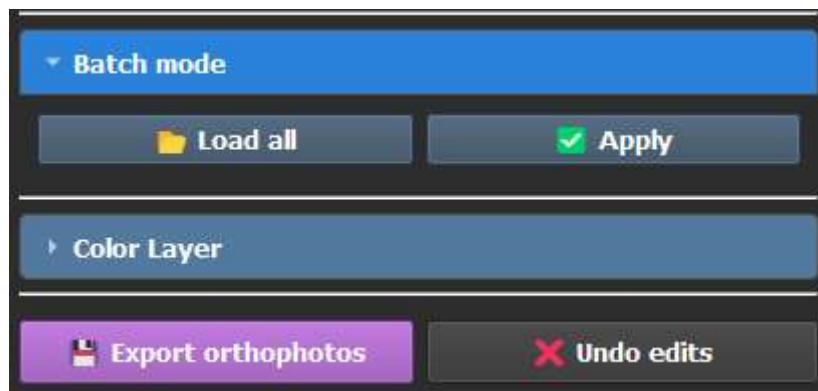
Black and White thresholds : exclude pixels which RGB mean is lower or upper configured value. White threshold is set by default to 220 in order to preserve white balance for snow.

Apply : when you change color settings for a selection, depending on its size, a downgraded version of the color layer over the selection is created, so that the slider moves have immediate effect on screen. Apply button makes the changes to the actual color layer. **SAVE BEFORE APPLY.**

Cancel : undo all changes that were made with the sliders.

Reset : reload the original colors from the JPEG files, restricted to the current selection. This feature is useful when the map has many selections already applied, and you want to get back to one of these selections in order to rework its color settings, without having to reset the whole color layer to its original state.

8.3 Batch mode



starting from the raw color layer, **Load all** the selections and their color settings from the .JSON file, and then **Apply** all the color settings.

Scenario :

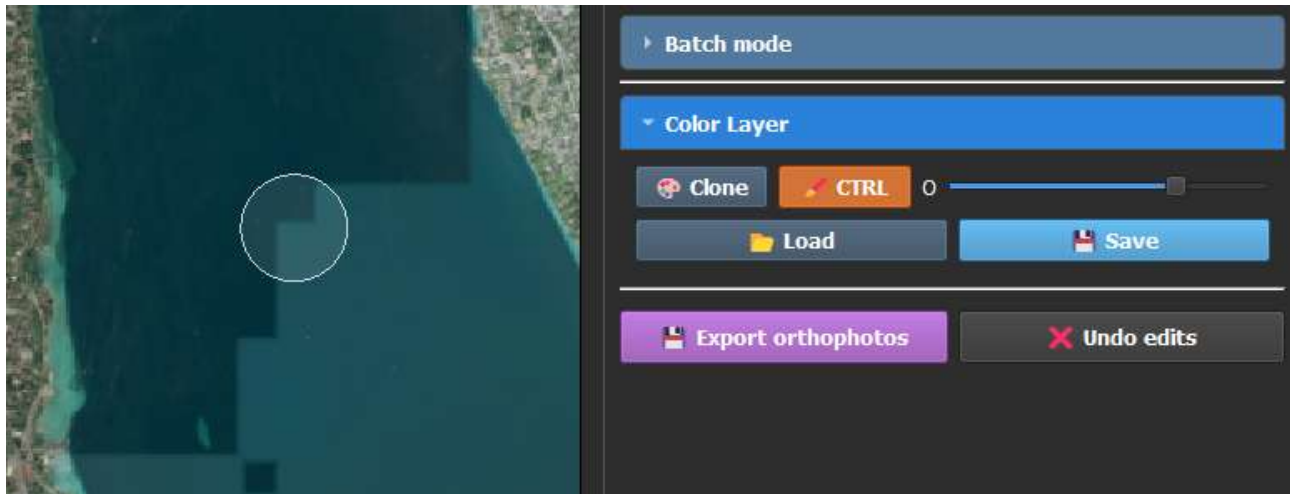
1- sometimes, creating multiple complex selections and their color settings may take a few hours for a single ortho tile. You can save your work at any time and come back to it afterwards, starting with the batch mode load.

2- when you are done with all the selections and their color settings, click on « Undo edits » button > confirm > Batch mode > Load all > Apply. Finally, one of the selections looks weird ? Selection > Load > Color adjustments > Reset > Load settings... and then you can slightly adjust the sliders before to save and apply the color settings.

3- someone has created and shared .JSON files containing the selections and their color presets. Load them in batch mode and you can create the updated orthophotos without having done anything by yourself.

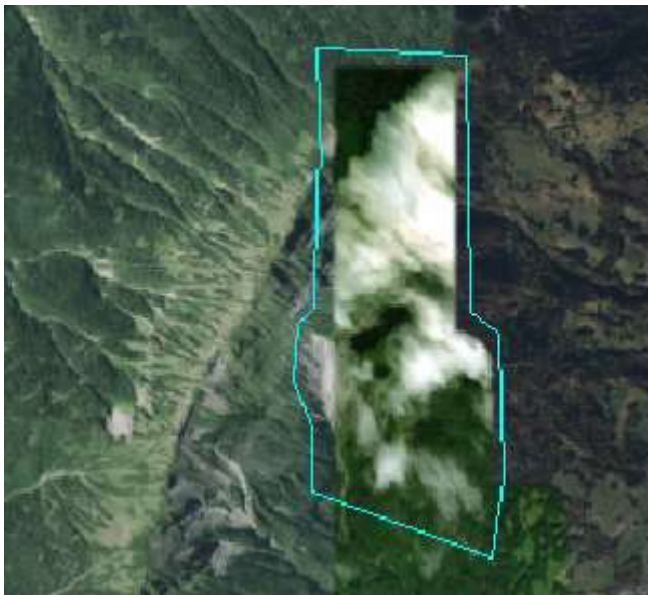
4- after loading all the selections together, each one appears as a red mask. The Apply button will show the final retouched colors. Therefore, it is possible, before to apply, to find out mistakes. Area not covered by a selection = terrain color shows up between red masks. Area covered by 2 selections = shows up with dark blue mask. However it may not matter in some cases, as long as we don't have 2 color settings that will apply on the same set of pixels.

8.4 Color Layer



Blur :

Click on button, adjust the slider, and CTRL + drag to blur an area. Useful on water.
This change cannot be saved to a .JSON file.



Clone :

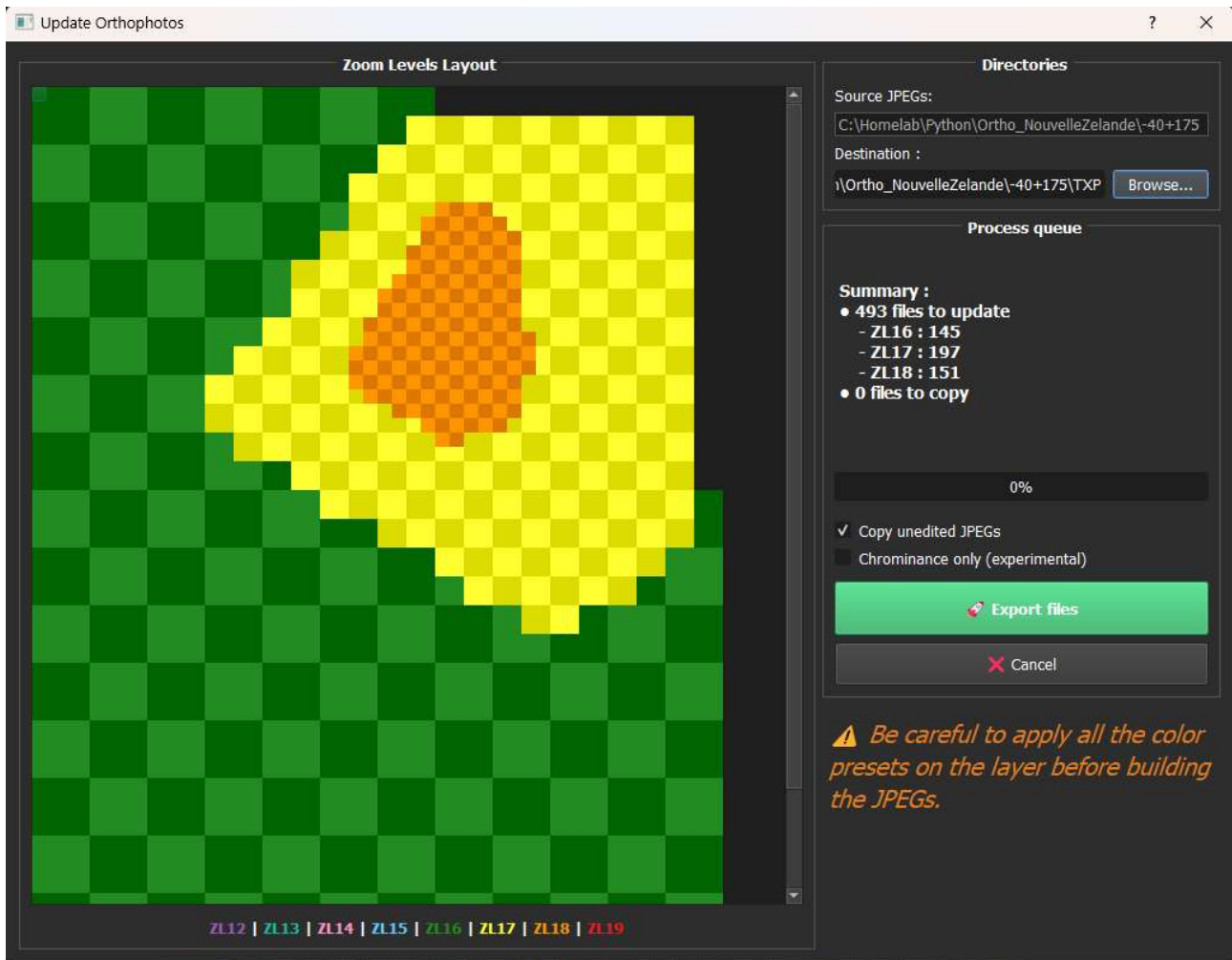
create or load a selection created for the cloning (no color settings). Typically, a cloud printed on the terrain. The selection must be slightly larger than the cloud. Click on « Clone » and then CTRL + drag the selection to another place, for example a place showing forest only. Release the mouse button and the cloning is done.

Load : load a color layer that was previously saved.

Save : save the current color layer, including the blur / clone effects.

A color layer backup file can be used afterwards to generate the new orthophotos.

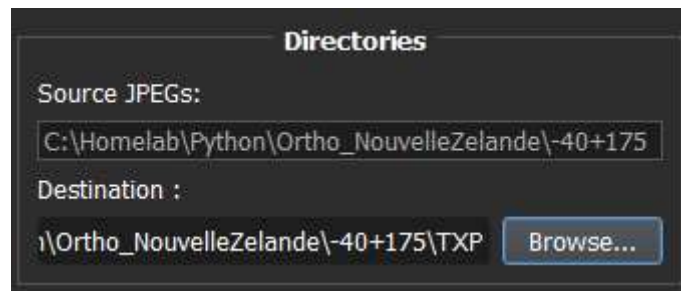
8.5 Export Orthophotos



The schema on the left shows the layout of the JPEG files that need to be updated according to the changes that were made and applied on the color layer.

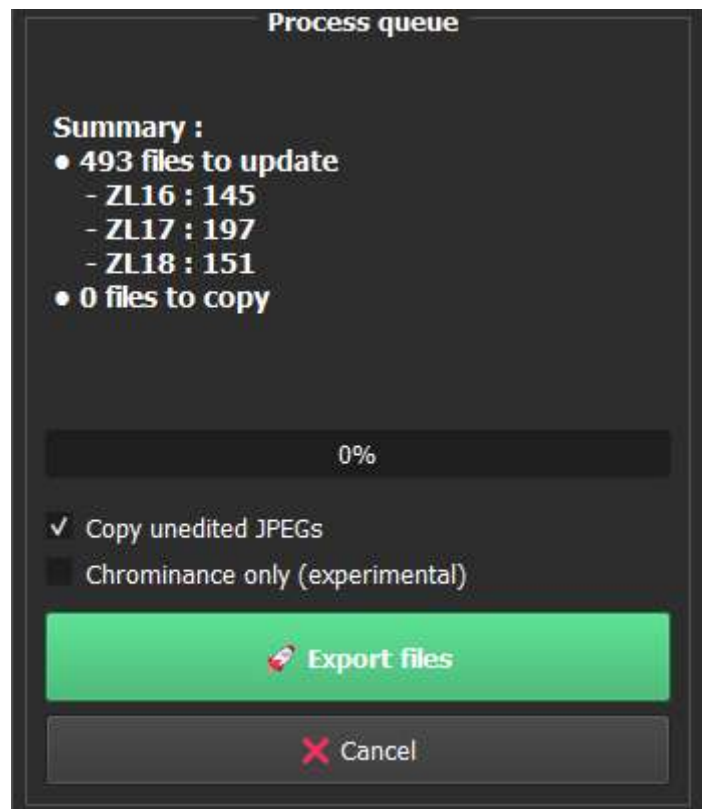
In most cases, the color layer size is much smaller than the original pictures. The JPEGs are modified by using a projection of the delta-RGB matrix from the color layer, so that the original details are preserved.

When a single JPEG file is crossed by a border between 2 selections, one side will receive color settings #1 and the other side will receive #2, so that the result in X-plane will be as nice as your skill to create accurate selections.



Source JPEGs : not editable. Refer to Mesh tab.

Destination : by default, will create a sub-folder « TXP » where the new orthophotos will be exported. This can be changed.



Copy unedited JPEGs : checked by default. Will copy all the JPEG files that are not modified with new color settings into the new directory.

Chrominance only : experimental feature for applying a color layer created from a lower ZL tile that has better colors than the higher ZL tile that we are currently working on. Results are questionable in detailed areas like cities. As an example, avoid applying a ZL12 color layer onto a ZL17 tile.

Export files : multi-threaded process to create the JPEGs. The degree of parallelization is calculated depending on the RAM available and the number of threads of the CPU. The duration can be up to a few minutes depending on tile ZL's and PC performance.

Cancel : close window.

9. Post-actions in Ortho4XP

9.1 Only mesh was altered

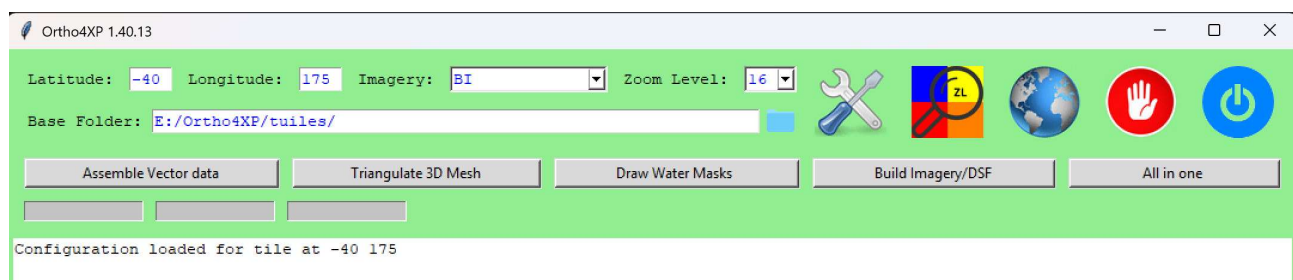
Earth nav data	28/06/2026 21:41
terrain	28/06/2026 21:41
textures	28/06/2026 21:42
Data-40+175.alt	22/02/2026 20:29
Data-40+175.apr	22/02/2026 20:29
Data-40+175.mesh	30/05/2026 10:18
Data-40+175.node	22/02/2026 20:29
Data-40+175.poly	22/02/2026 20:29
Data-40+175_orig.mesh	23/02/2026 18:49
Ortho4XP_-40+175.cfg	30/05/2026 10:27
Ortho4XP_-40+175.cfg.bak	29/05/2026 23:09

Typically, a new mesh file was exported with an alternate name.

Rename original mesh file like above (2).

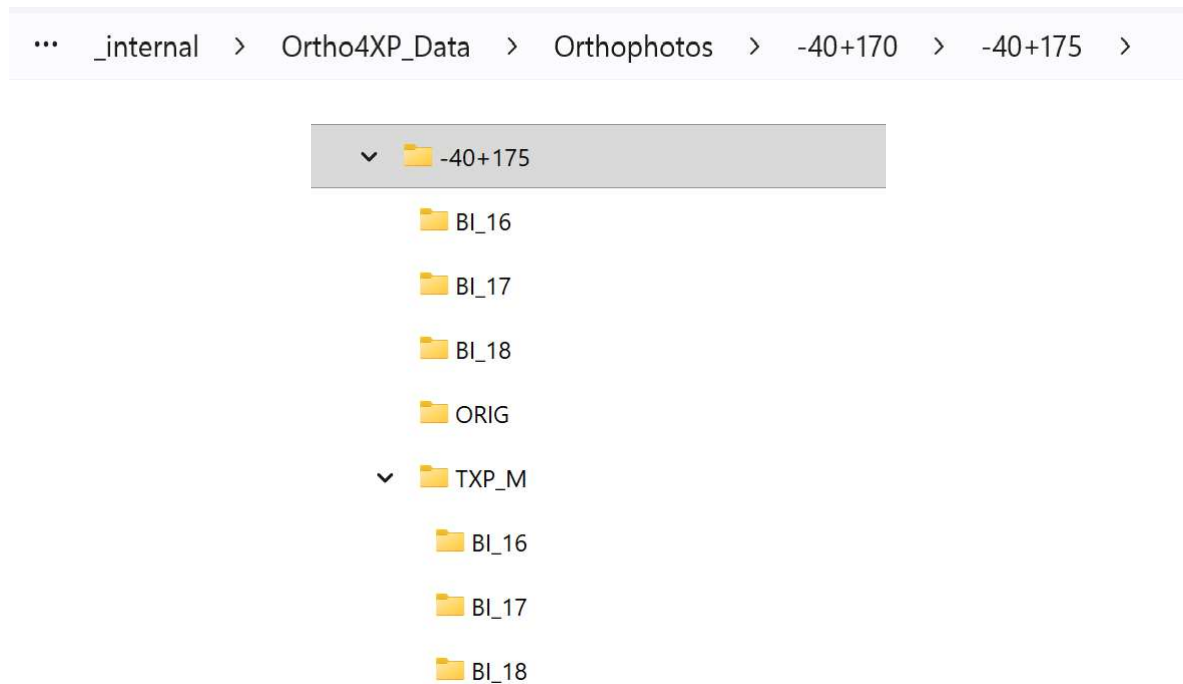
Then **rename new mesh file** so that Ortho4XP will use it (1).

Run Ortho4XP and select the right tile.



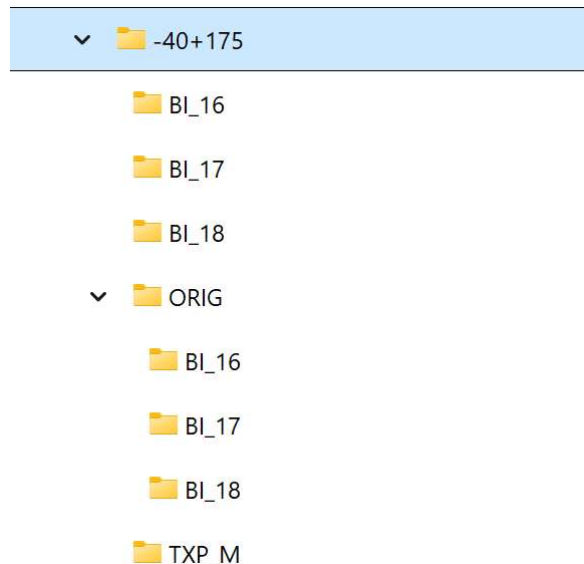
We are in a state where the mesh was just created. Click on « **Draw Water Masks** », wait for completion. Then click on « **Build Imagery/DSF** ». Only the .dsf file will be rebuilt, this is what we need. The .DDS will remain unchanged.

9.2 Only orthophotos were altered



After exporting the orthophotos, a new folder is created with the modified JPEG files. In this example, it is « TXP_M ».

Create a new folder « ORIG » (or any other name) and **move the original JPEG files** into it. Then **move the sub-folders from « TXP_M » to the upper level** of the directory tree.

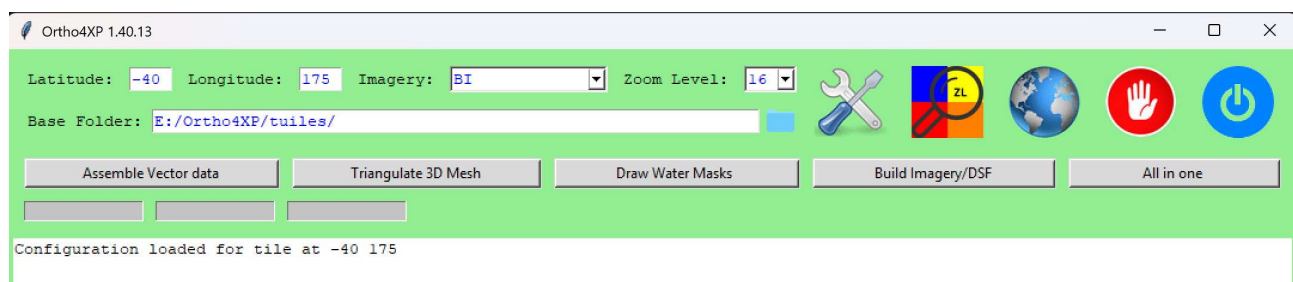


This is the expected result.

Earth nav data	28/06/2026 21:41
terrain	28/06/2026 21:41
textures	28/06/2026 21:42
Data-40+175.alt	22/02/2026 20:29
Data-40+175.apr	22/02/2026 20:29
Data-40+175.mesh	30/05/2026 10:18
Data-40+175.node	22/02/2026 20:29
Data-40+175.poly	22/02/2026 20:29
Data-40+175_orig.mesh	23/02/2026 18:49
Ortho4XP_-40+175.cfg	30/05/2026 10:27
Ortho4XP_-40+175.cfg.bak	29/05/2026 23:09

Now open the ortho tile folder and **delete « terrain » and « textures » sub-folders**.

Run Ortho4XP and select the right tile.



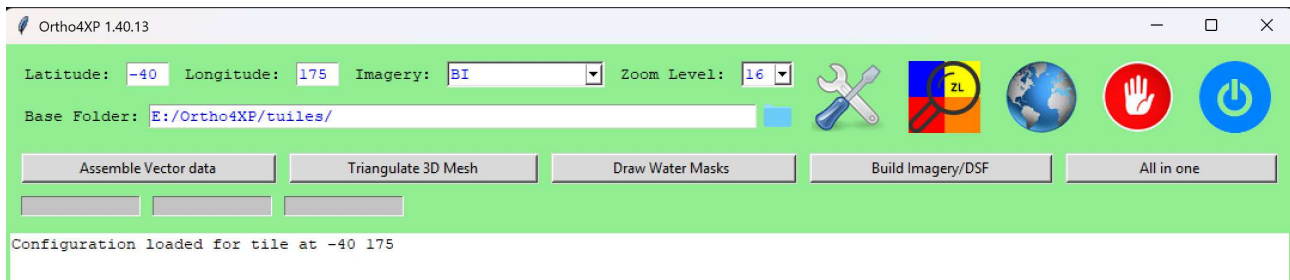
Click on « **Build Imagery/DSF** ». The .dsf file and all the .DDS will be rebuilt from the new JPEGs.

9.3 Mesh and orthophotos were altered

In this case, combine the actions in the directories of the tile and the orthophotos, like seen before :

- rename mesh files
- delete « terrain » and « textures »
- re-arrange the orthophotos directories.

Run Ortho4XP and select the right tile.



We are in a state where the mesh was just created and the imagery is missing. Click on « **Draw Water Masks** », wait for completion. Then click on « **Build Imagery/DSF** ». The .dsf file and all the .DDS will be rebuilt from the new JPEGs.

10. File management

10.1 Custom Json files

These files are created / populated when the « Save » buttons are used in the UI.

 +27+086_custom_2D_selections.json

 +27+086_custom_colors.json

 +27+086_custom_flat.json

 +27+086_custom_runways.json

custom_flat : surfaces to be flattened

custom_runways : runways to be created

custom_2D_selections : selections made in the « Colors » tab

custom_colors : color settings for the selections

relief_presets.json : smooth / chaos presets.

Having these files shared on the internet allow to process mesh and orthophotos changes without spending time on designing.

10.2 Settings.ini

This file is loaded at startup of TrianglesXP, and automatically saved right before the app is closed.

It contains parameters that you can change and will apply at next run of the app.

```
[Window]
maximized = False
width = 1612
height = 980
splitter_sizes = 1177,431
font_size = 10

[Lang]
language = en

[Paths]
texture_dir = C:\
last_mesh_dir = C:\
dds_dir = C:\
```

font_size : change the value of the characters size in the UI

```
[System]
max_history = 10
max_textures = 4
```

max_history : undo/redo stacks

max_textures : maximum number of HD textures that can be displayed together in 3D mode

```

[Tools]
max_subdiv_level = 4
bank_z_freeze_mask = 19
flatten_bank = 100
zone_subdiv = 1
runway_max_pts = 7|
runway_width = 60
runway_max_width = 500
runway_bank_side = 100
runway_subdiv = 1
runway_rect_size = 10
runway_type = 8

```

max_subdiv_level : maximum number of triangle subdivisions allowed within 1 operation on mesh
bank_z_freeze_mask : mask to protect the altitude of triangles based on their types. 19 = 1 (water) + 2 (sea) + 16 (runway).

runway_max_points : maximum points that can be created to define the shape of a new runway.

runway_rect_size : width of the triangles created to build a runway (do not change).

runway_type : triangle type of a new runway. It is 8 by default in OSM patches, so it's the same here. Can be changed to 16.

The most of the other parameters are for pre-configured values in editable fields in the UI.